

While loops in Python

- * Sometimes we need to repeat a task multiple times
- * A while loop repeats a block of code as long as the condition is True

Syntax:

```
while condition:  
    statement
```

eg:

```
print(1)  
print(2)  
...  
print(10)
```



```
i = 0  
while i <= 10:  
    print(i)
```

Note:

- * If the condition never becomes False, the loop runs forever
- * Always update variable inside the loop

Quiz:

1. Write a program that prints odd numbers from 1 to 20
2. Write program that prints numbers in reverse order

For loop in Python

* A for loop is used to repeat a block of code multiple times

* Unlike the while loop, we usually use a for loop when we know how many times we want to repeat something

Syntax:

```
for variable in sequence  
    statement
```

using range()

```
for i in range(start, stop)  
    statement
```

Note:

step: used to decide how much the numbers increase

Questions:

1.

```
for i in range(3):  
    print(i)
```

a. 123

b. 012

c. 0123

d. Error

2. Print multiplication table of 4

break, continue and pass

→ break:

The break statement is used to exit the loop immediately, even if the condition is still true.

Syntax:

```
for i in range(5):
```

```
    if condition:
```

```
        break
```

→ continue:

The continue statement is used to skip the current iteration and move to the next iteration.

Syntax:

```
for i in range(5):
```

```
    if condition:
```

```
        continue
```

→ pass

The pass statement does nothing. It is used as a placeholder when a statement is required syntactically.

Syntax:

```
for i in range(5):
```

```
    pass
```

Assignment:

1. Write a program that prints from 1 to 10 but stops when number = 7
2. Print numbers from 1 to 10 but skip multiples of 3

for i in range(1, 10):

if i % 3 != 0:

print(i)

print this pattern

* * *

* * *

* * *

(1,1) (1,2) (1,3)

(2,1) (2,2) (2,3)

(3,1) (3,2) (3,3)

Nested loops

=> A loop inside another loop

idea:

outer loop → runs first

inner loop → runs completely for each outer loop iteration

syntax:

```
for i in range(3):
```

```
    for j in range(3):
```

```
        print(i,j)
```

Assignment:

1. Print this pattern

```
* * *
```

```
* * *
```

```
* * *
```

2. Write a program that print all possible pairs of numbers

```
(1,1) (1,2) (1,3)
```

```
(2,1) (2,2) (2,3)
```

```
(3,1) (3,2) (3,3)
```

Pattern Problems

1. Write a program to print

1

1 2

1 2 3

1 2 3 4

2. Write a program to print

* * *

* * *

* * *

*

3. Write a program to print

1

1

2 3

2 2

4 5 6

3 3 3

7 8 9 10

4 4 4 4

4. Write a program to print

*

* *

* * *

* * * *

Loop Practice Questions

1. Write a program to print sum of n natural numbers
2. Write a program to find the factorial of a number
3. Write a program to reverse a number
4. Write a program to count the number of digits
5. Write a program to check whether the number is a palindrome